CII SR -15th Edition Online Kaizen Compétition 2020

A Kaizen Presentation By

(Everest Industries Ltd)

Podanur Works - Coimbatore

18th ~ 20th August 2020

Company Profile

COMPANY OVERVIEW

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

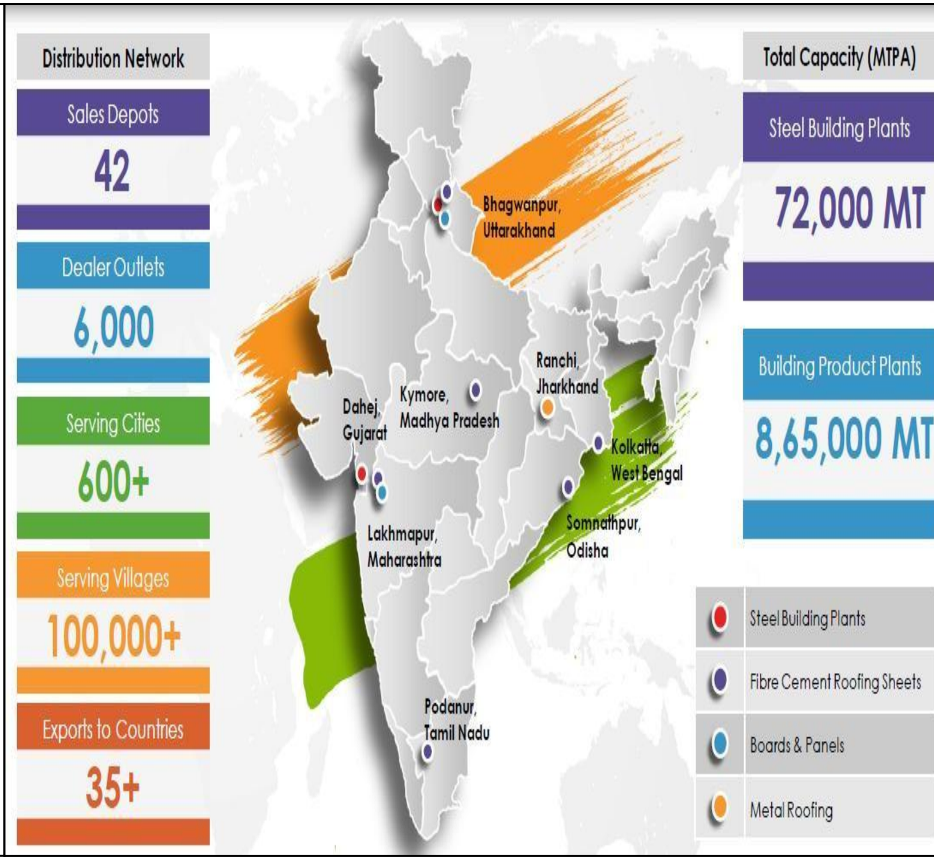
OUR BUSINESS

The company businesses are divided into two distinct business segments
with capacity –

1. Buildings Products – 8,65,000 MT / Annum
2. Steel Buildings – 72,000 MT / Annum

Buildings Products segment contributed 63% of the company's revenue and includes Fiber cement roofing sheets, Fiber cement boards, Rapicon wall & their accessories.

Steel Building segment accounted for 37% of annual revenue. In this segment Company offers customized building solutions in steel such as Pre-Engineered steel Buildings, Smart steel buildings, Metal roofing and Cladding.



Company Products & Customers

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Products:


Fibre Cement Roofing Sheets


Hi-Tech Roofing Sheets


Super Sheets


Cement Planks


Fibre Cement Boards


Rapicon Walls


Smart Steel Buildings

Customers:

L&T
Reliance Industries Limited
Simplex Infrastructures Limited
Walmart Save money. Live better.
IFB
ACC
Bentley
Mahindra
HYATT CAPITAL GATE ABU DHABI
Pidilite
TATA
Indian Railways
Marriott HOTELS & RESORTS
ITC Limited
Whirlpool CORPORATION
RELIANCE
BRITANNIA Eat Healthy, Think Better
DLF
DLF BUILDING INDIA
NBCC A Navratna CPSE
Cipla
Coca-Cola
VIDEOCON

Kaizen Theme

Kaizen Theme: To improve productivity by reducing template supply cycle time to Machine.

Implemented Area: Line No - 2, Podanur Works, Coimbatore

Implementation Date: 02nd August 2020

Cross Functional Team:

Mr. A.N. Vignesh
(Engineer - Maintenance)

Team Leader
Mr. M. Vijay Kumar
(Senior.Engineer - Electrical)

Mr. K. Senthil Kumar
(Electrical Supervisor)

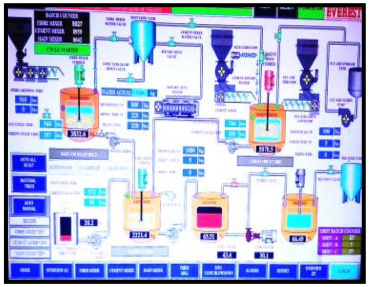
Mr. A. Devaraja
(Fitter)

Mr. Pillai
(Fabricator)

Mr. M. Srinivasagam
(Electrician)

Problem / Present Status

Process Flow:

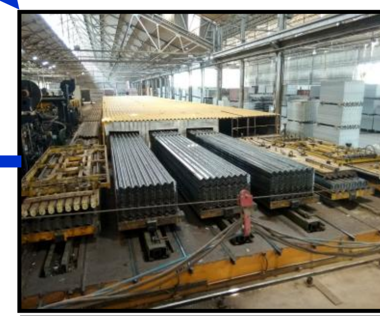


1) RAW MATERIAL PREPARATION

2) SHEET FORMING

3) SHEET HANDLING/ PILING MACHINE

SHEET RECYCLING UNIT



4) PRE-CURING TUNNEL

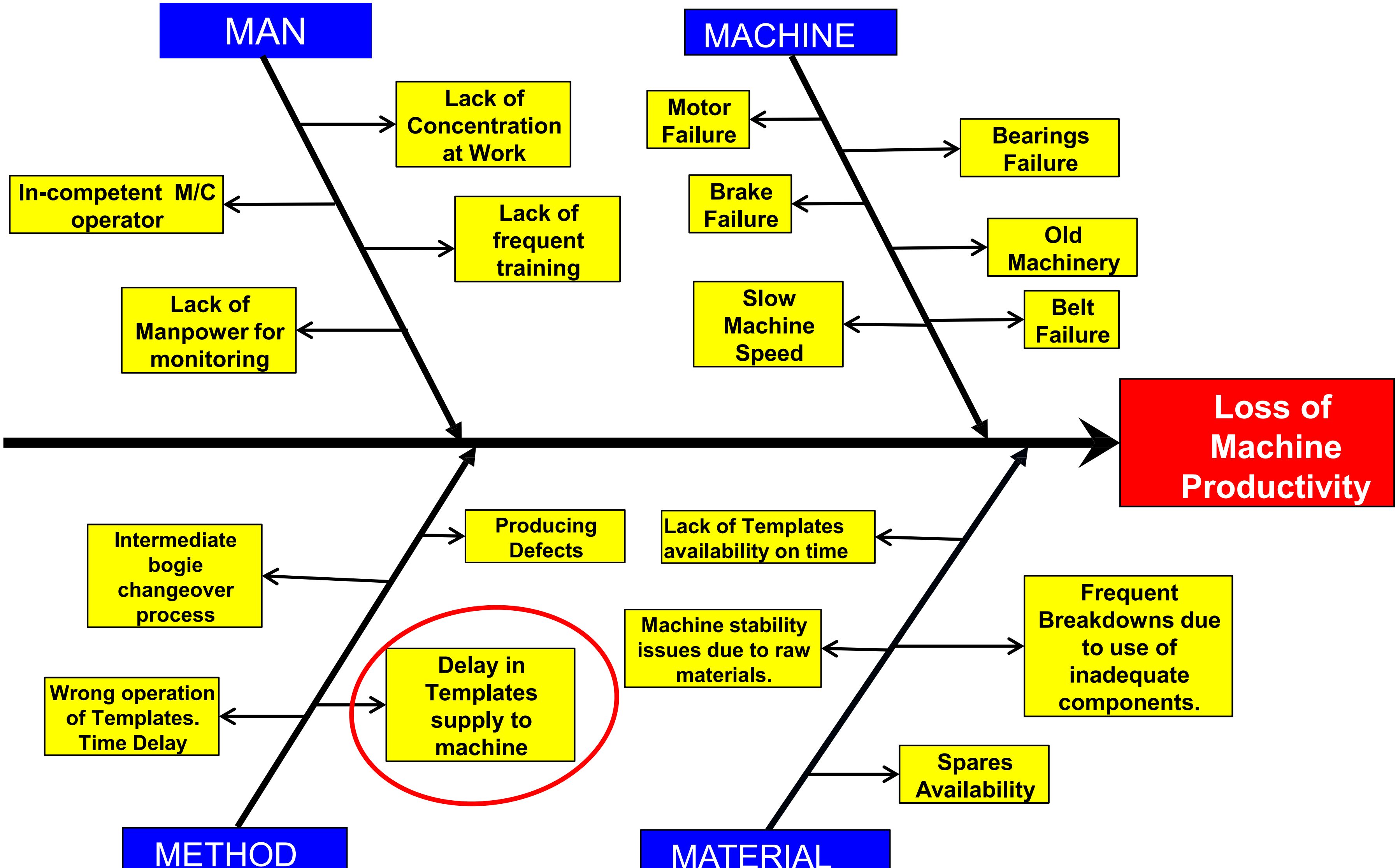
5) DE-PILING MACHINE

6) TEMPLATE PILING MACHINE

Problem/ Present Status :

Loss of Machine Productivity.

Root Cause - Cause & Effect Analysis



Root Cause - Why Why Analysis

Problem Statement	Delay in templates supply to machine.
Why 1	Due to templates stacker machine.
Why 2	Template stacker pad has to be lowered to stack each & every template in bogie.
Why 3	No provision to lift template bogie.

Root Cause:
No provision to lift template bogie.

Idea Generation: To lift template bogie by using fork-lift mechanism.

Action:
To provide hydraulic lifting mechanism to lift the template bogie.

Countermeasure

S.No	Counter Measure Activities	Start Date	Target Date	Responsibility	Status
01	To design bogie lifting arm arrangement for lifting 10 M.T Bogie.	03/02/2020	27/02/2020	Mechanical & Electrical Department	Completed
02	To use scrap hydraulic cylinder & modify it's lifting capacity to 10 M.T by cutting cylinder to reduce it's stroke length.	28/02/2020	25/03/2020	Mechanical Department	Completed
03	To do Civil work in bogie passing track & to cut small piece of Rail and weld it to bogie lifting arm for matching the track.	04/07/2020	10/07/2020	Mechanical Department	Completed
04	To make logic through PLC & it's working fully automatically as per the logic. And to make interlock for safety purpose.	11/07/2020	15/07/2020	Electrical Department	Completed

Before Situation

❖ Previously, there was no mechanism to lift template piling bogie. So, templates stacking pad has to be lowered for each and every time for stacking templates in the bogie. Due to this, there is delay in template supply to machine. And hence, to prevent any sheet loss due to template delay, we are reducing the machine speed to 16 seconds/sheet cycle.

Before Situation

❖ Previously, there was no mechanism to lift template piling bogie. So, templates stacking pad has to be lowered for each and every time for stacking templates in the bogie. Due to this, there is delay in template supply to machine. And hence, to prevent any sheet loss due to template delay, we are reducing the machine speed to 16 seconds/sheet cycle.

After Situation

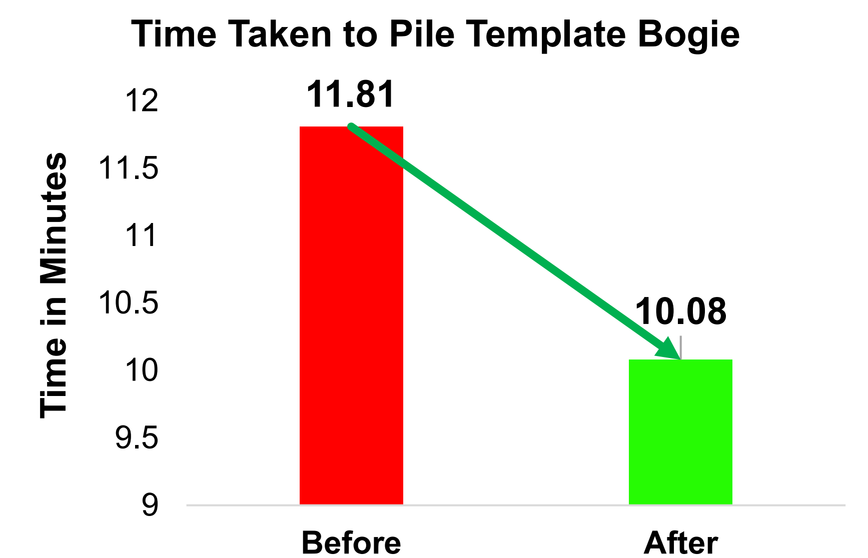
❖ Now, template bogie lifting mechanism lifts the bogie by using fork lift mechanism. Due this, there is no need to lower templates stacking pad each & every time for stacking templates in bogie. By this templates supply time is matching with machine speed & there is no loss of productivity due to delay in templates supply.

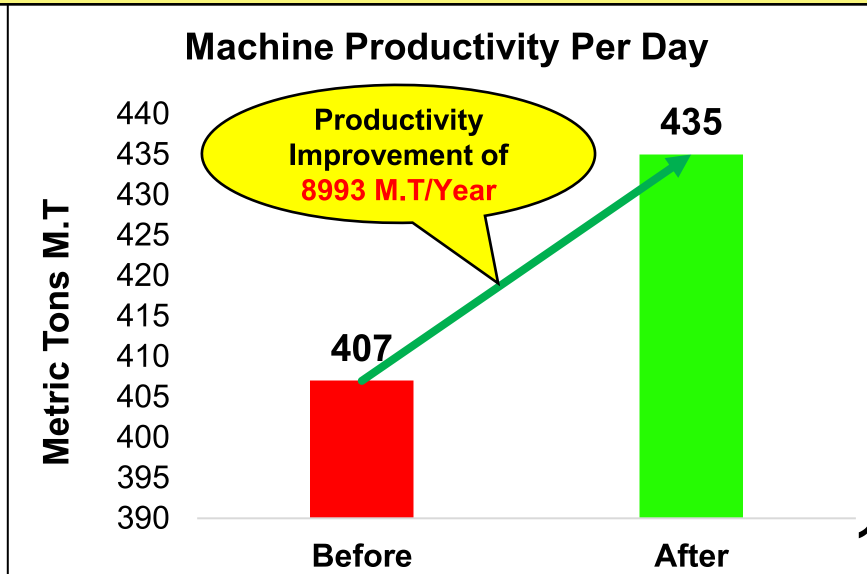
After Situation

- Now, template bogie lifting mechanism lifts the bogie by using fork lift mechanism. Due this, there is no need to lower templates stacking pad each & every time for stacking templates in bogie. By this templates supply time is matching with machine speed & there is no loss of productivity due to delay in templates supply.

Validation of Data

Template Bogie Piling Data		
	Before	**After**
Time taken by Template pad to pile 1 Template in bogie	12.89 sec	11 sec
Time taken by Template pad to pile 1 bogie (55 Templates)	11.81 minutes	10.08 minutes
Machine Running per sheet cycle	16 sec	14.5 / 15 sec
Total Sheets produced in a Day	5400 Sheets/Day	5760 Sheets/Day
Total M.T produced in a Day	407 M.T/Day	435 M.T/Day
Productivity improvement / Day	28 M.T / Day	
Productivity improvement / Year	**8993 M.T / Year**	





Benefits Achieved

Tangible Benefits

- Productivity of the Machine has increased by reducing template supply cycle time to machine.
- Template supply cycle time is matching with machine speed.
- No damage to templates profile while dropping them on bogie.
- Template stacker pad up-down movement motor failure is eliminated.
- Less investment cost of INR. 20,000/- instead of spending 8 Lakhs for CLT Project by using scrap parts(In-house Project).
- Reduction in oil spillages.

In-tangible Benefits

- Morale of our People increased due to increase in Productivity.
- Sound levels while stacking templates in bogie has reduced.

Standardization & Horizontal Deployment

- Training on Updated Work Instructions are given to all the workmen for regular inspection & maintenance.

EVEREST	EVEREST INDUSTRIES LIMITED PODANUR WORKS IS/ISO 9001 : 2015 WORK INSTRUCTION	DOC NO :EIL/WI/512 ISSUE NO : 01 REV NO : 01 PAGE : 1 OF 1 DATE : 01/02/2018

Work Instruction: 512

Template Piling Operation

Sequence of Operation:

- Templates are supplied to the Template Piling machine through roller conveyor from De-piling machine.
- Template bogie which is in Template piling machine is lifted by using huydraulic cylinders upto the level of template pad
- Templates are picked up from roller conveyor by Templates piling pad & piles them in bogie.
- Template piling pad position is need not to be lowered, as the bogie level is increased till the piling pad.
- Once the Template bogie is filled, change it to new bogie.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection & Maintenance.
HOW TO DO	Regular Training on Standardized work instructions
FREQUENCY	Weekly

Horizontal Deployment:

BW	CW	KW	LW	PW	SW

- Under Progress - Deployed

Kaizen Sheet

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Kaizen Sheet (everest)							Company	MM/YY	S.No
Productivity	Quality	Delivery	Safety	Material	Energy	Maintenance	Everest Industries Ltd	07/20	1

Kaizen Title: To improve productivity by reducing template supply cycle time to Machine.

Implemented Area:
Shopfloor Line no.2

Problem/Present Status:
Loss of Machine productivity.

Before Improvement:

Implemented by:
M.Vijay Kumar, A.N.Vignesh,
K.Senthil Kumar, Pillai,
A. Devaraja, M. Srinivasagam

Result/Benefit:

(a) Qualitative
Productivity of the Machine has increased by reducing template supply cycle time to machine.

Root Cause Identification:

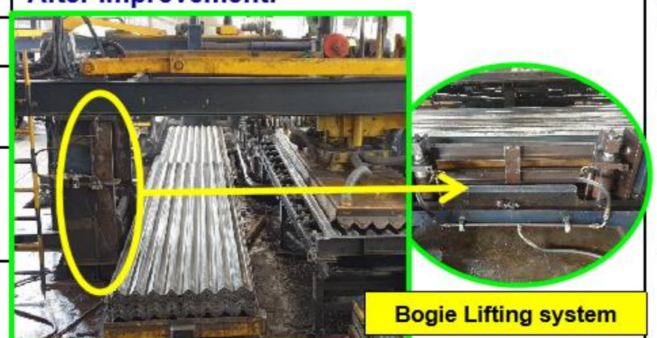
Why 1: Due to reduced Machine speed.

Why 2: Delay in Templates supply to Machine by templates stacker.

Why 3: Template stacker pad has to be lowerd to stack each & every template in bogie.

Why 4: No provision to lift template Bogie.

After Improvement:



(b) Quantitative
- Template supply cycle time is matching with Machine speed.
- No damages to templates profile while dropping them on bogie.
- Template stacker pad up-down movement motor failure is eliminated.
- Less Investment cost of 20,000/- due to in-house built, instead of spending 8 Lakhs for CLT project.

Root cause	No provision to lift template bogie.
Idea to eliminate root cause	To lift Template bogie up to required height.
Counter-measure	Provided Hydraulic lifting mechanism to lift the template bogie. By this, there is no need to lower template stacker pad for stacking templates in bogie. Due to time template supply cycle time is reduced.

Standardization:
Training is provided on SOP to workmen.

How many places this Kaizen can be deployed horizontally

BW	CW	KW	LW	PW	SW

CII

Thank You